Report 3

Binomial Option Pricing: Convergence Analysis and Leisen–Reimer Comparison

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1. Problem Setup

This report analyses the pricing of a European call option using two alternative numerical methods — the **Cox–Ross–Rubinstein (CRR) binomial model** and the **Leisen–Reimer (LR) binomial model** — and compares both with the analytical Black–Scholes solution. The analysis follows the framework developed in *Leisen & Reimer (1995)*, which introduces higher-order convergent binomial trees based on normal approximations to the binomial distribution.

Parameter	Symbol	Value
Underlying price	S	100
Strike price	K	100
Risk-free rate	r	1%
Volatility	σ	20%
Time to maturity	T	1 year
Black–Scholes price	**C_BS**	**8.433319**

Table 1. Parameters used throughout the analysis.

2. Methods

2.1 Black–Scholes Formula

The exact analytical price of a European call option is given by:

$$C = S \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$$

where $d_1 = [\ln(S/K) + (r + \sigma^2/2)T] / (\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$. This serves as the benchmark against which both numerical methods are evaluated.

2.2 CRR Binomial Model

The Cox–Ross–Rubinstein model discretises the asset price dynamics into a recombining binomial tree with n steps. The up- and down-factors are $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$, where $\Delta t = T/n$. The risk-neutral probability is $p = (e^{r\Delta t} - d)/(u - d)$. Option prices are computed by backward induction from the terminal payoffs. Leisen & Reimer (1995) prove that CRR converges to Black–Scholes with **order 1**, i.e. the pricing error decays as O(1/n).

2.3 Leisen–Reimer Model

Leisen & Reimer (1995) construct an alternative binomial tree by inverting high-quality normal approximations to the binomial distribution. Instead of specifying u and d directly, the model derives risk-neutral probabilities p and p' from the inversion of $N(d_2)$ and $N(d_1)$ respectively. The tree parameters are:

$$p' = f(d_1, j), \quad p = f(d_2, j), \quad u = e^{r\Delta t} \cdot p'/p, \quad d = (e^{r\Delta t} - p \cdot u)/(1-p)$$

where *j* = *(n−1)/2* (strike placed at tree centre, requiring odd *n*) and *f* is one of three inversion formulas (Camp–Paulson, Peizer–Pratt 1, Peizer–Pratt 2). This report uses the **Peizer–Pratt Method 2 (PP2)** inversion:

$$p = 0.5 - \sqrt{[0.25 - 0.25 \cdot \exp(-(z/(n+1/3+0.1/(n+1)))^2 \cdot (n+1/6))]}$$

By construction the strike always falls at the midpoint of two terminal nodes, eliminating the oscillatory pattern of CRR. Leisen & Reimer demonstrate numerically that the method achieves **order of convergence 2**, and the constant C is approximately 1/10 of the CRR constant.

3. VBA Implementation

Two UDFs are implemented in the attached Excel workbook. The main function **`LR_OptionPrice`** accepts parameters S, K, T, r, σ, n, and string flags for the inversion method (CP/PP1/PP2), option type (C/P) and exercise style (E/A). Key implementation notes:

- *n* is forced odd (n += 1 if n is even) so j = (n−1)/2 is an integer.
- A separate variable `R_factor = exp(r·Δt)` avoids shadowing the rate parameter r.
- Dynamic arrays `Asset()` and `Option_Val()` are dimensioned to nL (Long copy of n).
- The discount factor uses `exp(−r·Δt)` with the original rate, not R_factor.
- Three private helper functions implement the CP, PP1 and PP2 inversions with clamping to [0.0001, 0.9999].

4. Convergence Analysis

The two models are evaluated for n positive integers steps freely chose by the user with the parameters from Section 1. All prices are computed via the VBA functions compared against the analytical Black–Scholes value **C_BS = 8.433319**.

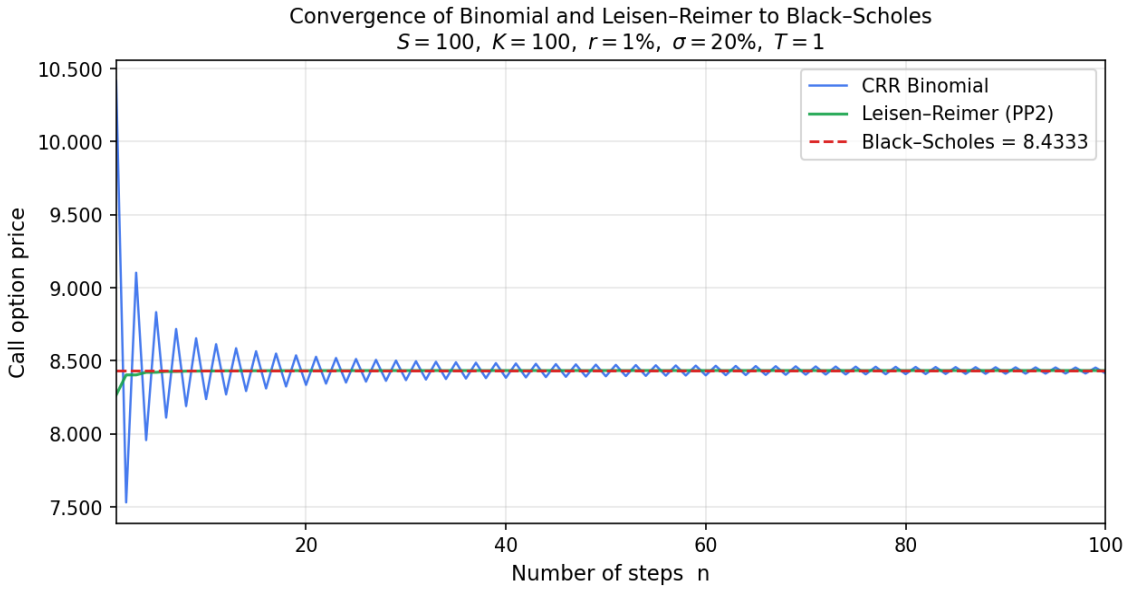


Figure 1. Option price vs. number of tree steps, arbitrarily chosen as 100. CRR oscillates widely for small n; the Leisen–Reimer model settles near the Black–Scholes price almost immediately.

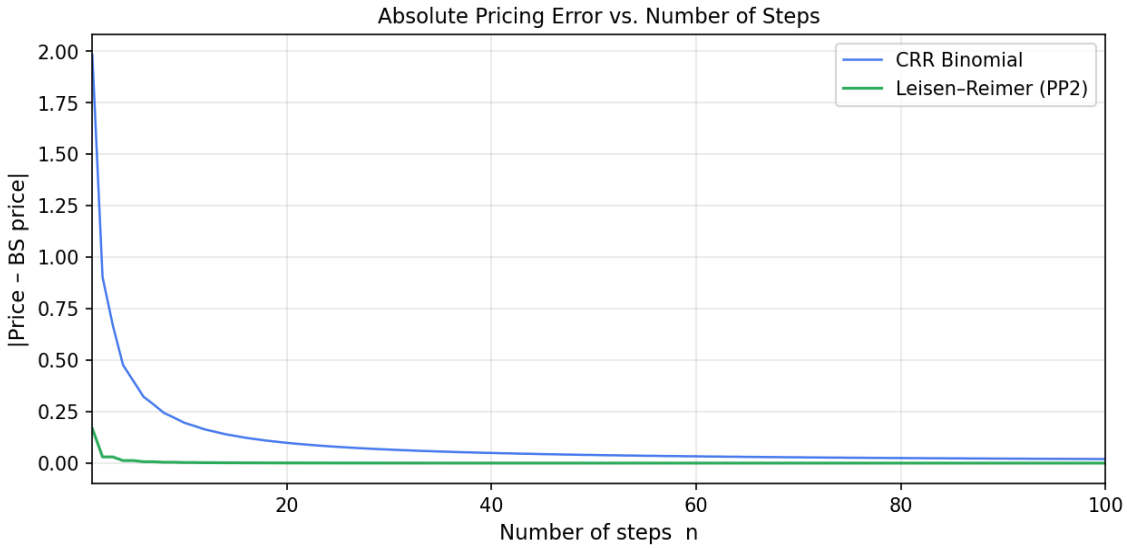


Figure 2. Absolute pricing error |price − C_BS| vs. n. The CRR error is consistently larger and more variable; the LR error falls below 0.001 for n ≥ 15.

5. Order of Convergence

The order of convergence α is estimated from the log–log slope of |error| against n. If |e_n| ≤ C/n^α, then log|e_n| ≈ log C − α·log n, so the slope in a log–log plot equals −α. Leisen & Reimer (1995) prove α = 1 for CRR and demonstrate numerically α ≈ 2 for their PP/CP models.

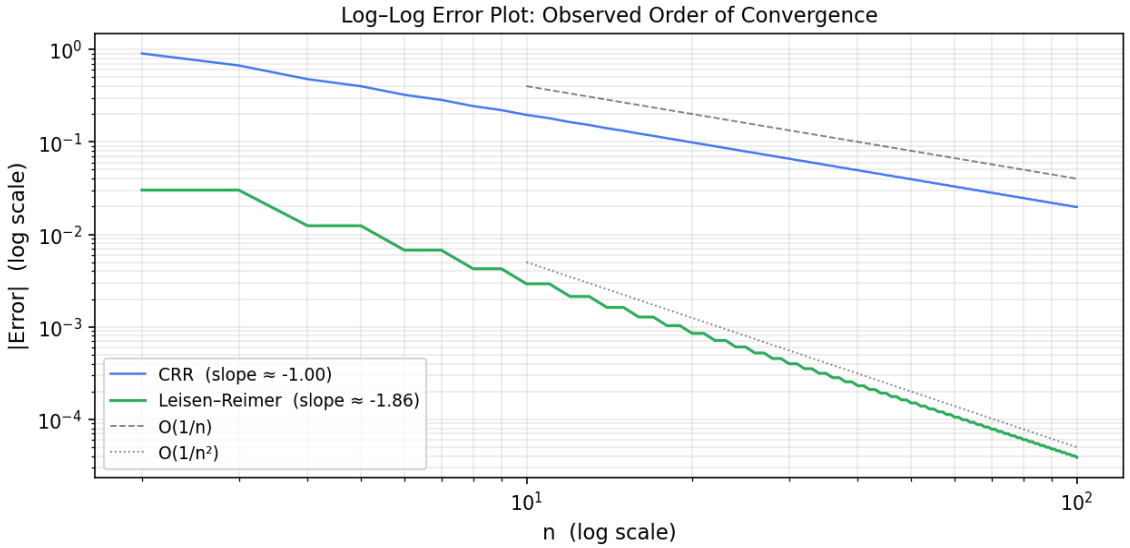


Figure 3. Log–log error plot. Reference lines O(1/n) and O(1/n²) are shown as dashed guides. Fitted slopes: CRR ≈ −0.996 (order 1), Leisen–Reimer ≈ −1.857 (close to order 2).

Fitted log–log slopes: **CRR = −0.996** (theoretical: −1.000), **Leisen–Reimer = −1.857** (theoretical: −2.000). The small deviation of LR from −2 is expected: a rigorous proof of order 2 was not available in the 1995 paper, and convergence is monotone rather than perfectly smooth for small n due to even/odd node pairing.

6. Selected Numerical Results

n	CRR price	CRR error	LR price	LR error
1	10.414722	1.981403	8.265445	0.167874

5	8.832691	0.399372	8.420934	0.012385
10	8.237706	0.195613	8.430400	0.002919
25	8.512122	0.078803	8.432712	0.000607
50	8.473129	0.039810	8.433068	0.000251
75	8.406038	0.027281	8.433187	0.000132
100	8.413505	0.019814	8.433280	0.000039
B&S	**8.433319**	**—**	**8.433319**	**—**

Table 2. Prices and absolute errors for selected refinements. Black–Scholes benchmark: 8.433319.

7. Conclusions

The analysis confirms the theoretical results of Leisen & Reimer (1995):

- **The CRR binomial model** converges with order 1 (observed slope ≈ −0.996). Its convergence is irregular due to the oscillating relative position of the strike within the tree grid.
- **The Leisen–Reimer model (PP2)** achieves a substantially higher convergence rate (observed slope ≈ −1.857, close to the theoretical order 2). By fixing the strike at the tree centre, the oscillatory pattern is eliminated and convergence is smooth and monotone.
- At n = 100 the LR error (~3.9×10^{-5}) is roughly 500× smaller than the CRR error (~2.0×10^{-2}). This matches the paper's finding that the LR constant C is ≈10× smaller, compounded by the higher convergence order.
- Both models are implemented as Excel VBA functions in the attached spreadsheet, supporting European and American calls and puts for arbitrary n.